



Mahindra University Hyderabad

Ecole Centrale School of Engineering

End-Semester Examination

Program: B. Tech. Branch: CAM Year: II Semester: I

Subject: Design and Analysis of Algorithms (MA 2107)

Date: 11/12/2025

Time Duration: 3 Hours

Start Time: 10:00 AM

Max. Marks: 100

Instructions:

1. There are 10 questions, all of which are compulsory.
2. Justify your response whenever necessary. Guesswork will not be considered in evaluation.
3. Answers without proper numbering will not be evaluated.

1. Answer the following questions:

10 M

a. Define $O()$, $\Theta()$ and $\Omega()$ notations.

6M

b. Solve the following recurrence relation:

$$T(n) = 2T\left(\frac{n}{2}\right) + n^2$$

4M

2. a. Arrange the following list of numbers in ascending order using the Merge Sort algorithm. Show all the steps.

10 M

8M

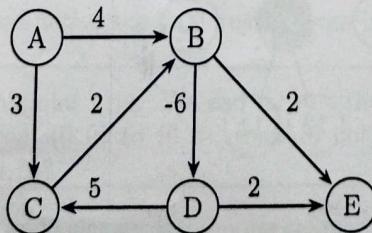
40, 17, 8, 2, 57, 32

b. What is the running time of Merge Sort?

2M

3. Consider the following graph:

10 M



- a. Define the subproblem for **All Pair Shortest Path** and show the relationship between them. 2M
- b. Find **All Pair Shortest Path** using **Floyd-Warshall Algorithm**. Show each DP table. 8M

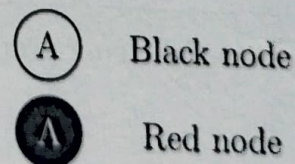
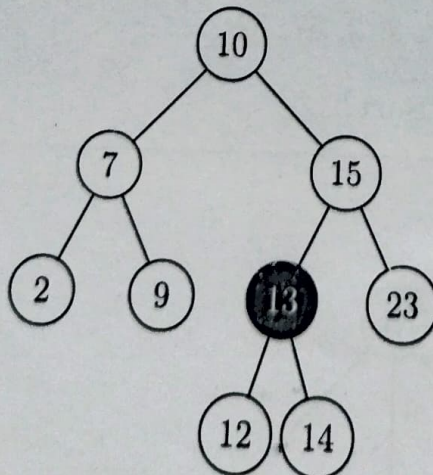
4. Using the given pseudo-code **RB-DELETE-FIXUP**, delete 2 from the given Red-Black tree and return a valid Red-Black tree. Show all the steps. 10 M

RB-DELETE-FIXUP(T, x)

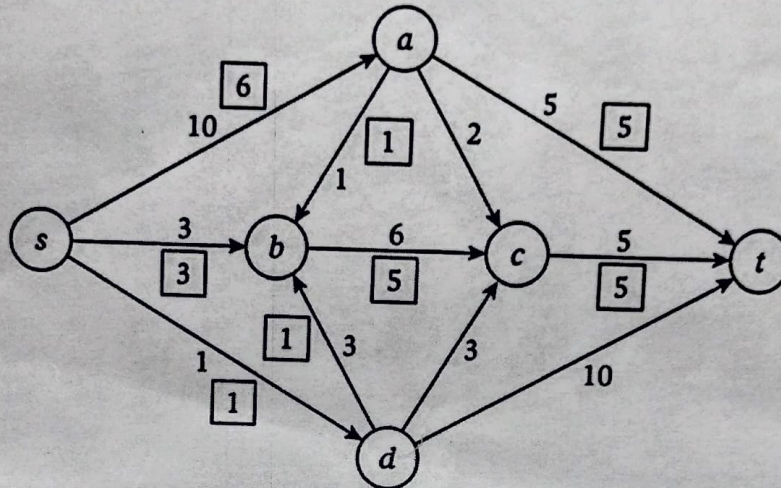
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1  while  $x \neq T.root$  and  $x.color == BLACK$ 
2      if  $x == x.p.left$ 
3           $w = x.p.right$ 
4          if  $w.color == RED$ 
5               $w.color = BLACK$  // case 1
6               $x.p.color = RED$  // case 1
7              LEFT-ROTATE( $T, x.p$ ) // case 1
8               $w = x.p.right$  // case 1
9          if  $w.left.color == BLACK$  and  $w.right.color == BLACK$ 
10              $w.color = RED$  // case 2
11              $x = x.p$  // case 2
12         else if  $w.right.color == BLACK$ 
13              $w.left.color = BLACK$  // case 3
14              $w.color = RED$  // case 3
15             RIGHT-ROTATE( $T, w$ ) // case 3
16              $w = x.p.right$  // case 3
17              $w.color = x.p.color$  // case 3
18              $x.p.color = BLACK$  // case 4
19              $w.right.color = BLACK$  // case 4
20             LEFT-ROTATE( $T, x.p$ ) // case 4
21              $x = T.root$  // case 4
22         else (same as then clause with "right" and "left" exchanged)
23      $x.color = BLACK$ 

```



5. a. Define maximum flow problem. 2M 12 M
 b. For a given flow f , define the corresponding residual graph G_f . 2M
 c. Consider the following flow network on which an $s - t$ flow has been computed. The capacity of each edge appears as a label next to the edge, and the numbers in boxes give the amount of flow sent on each edge. Edges without boxed number have no flow being sent on them. **Compute the residual graph with respect to the given flow.** 5M



- d. The given flow is a maximum flow. Prove or disprove. 3M

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6. a. Define **Maximum Interval Scheduling** problem. 2M 8 M
 b. Give a counter-example to show that **selecting interval with earliest start time** strategy can fail to produce an optimal solution for the Maximum Interval Scheduling Problem. 2M
 c. Define **Weighted Interval Scheduling** problem. 2M
 d. Give a counter-example to show that **selecting interval with maximum weight** strategy can fail to produce an optimal solution for the Weighted Interval Scheduling Problem. 2M

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7. Find the longest common substring/subword between **portable** and **passport** 10 M

- a. Write the subproblems and relationship among them. 3M
 b. Create the DP table and derive the final answer from it. 7M

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8. Roads are arranged in rectangular grid. We can move either up or right. How many different routes are there from $(0, 0)$ to $(6, 6)$, that do not pass through the blocked points at $(3, 2)$, $(2, 4)$ and $(4, 5)$? 10 M

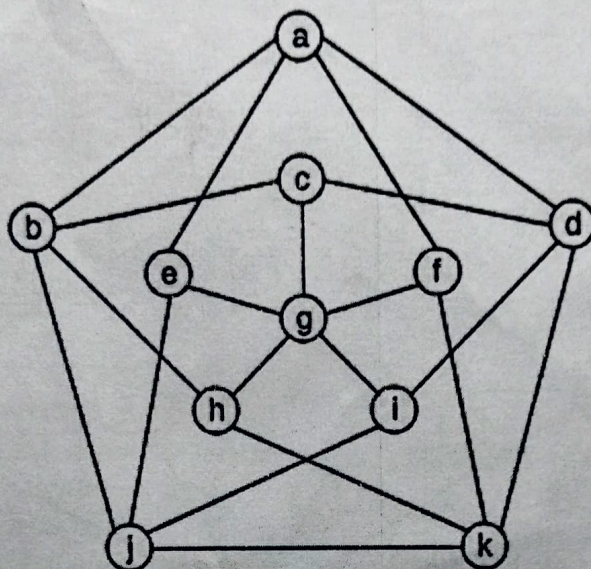
- a. Write the subproblems and relationship among them. 2M

b. Create the DP table and derive the final answer from it.

8M

9. Using Backtracking Algorithm check if the following graph is 3-colorable?

10 M



10. For a given graph $G = (V, E)$, a subset I of the vertex set V , is said to be an **Independent Set** if for any pair of vertices $u, v \in I$, $(u, v) \notin E$. 10 M

In, k -Independent Set problem, we ask, "Does there exists a k size independent set in G ?"

For a given graph $G = (V, E)$, a subset S of the vertex set V , is said to be a **Vertex Cover** if for any edge $(u, v) \in E$, either $u \in S$ or $v \in S$.

In, k -Vertex Cover problem, we ask, "Does there exists a Vertex Cover of size k in G ?"

Show that k -Independent Set is NP-complete. (Hint. you can use reduction from Vertex Cover.)
